



A Hybrid CNN–XG Boost Framework for Early-Stage Disease Diagnosis Using Multimodal Clinical and Imaging Data

Prof.Ms.J.Aiswarya¹,I.Lakshmi Krishna² and K.Saran³

¹Research Scholar, Department of Computer Applications, Sri Krishna Arts and Science College, Coimbatore.

^{2,3}Students III CSA, Department of Computer Applications, Sri Krishna Arts and Science College, Coimbatore.

Abstract : Early and accurate diagnosis of chronic diseases remains a critical challenge in clinical practice, where decisions must integrate heterogeneous data sources such as laboratory measurements, patient history, and medical images. This paper proposes a hybrid machine learning framework that fuses deep convolutional feature representations extracted from medical imaging data with structured clinical variables, and classifies patient outcomes using a gradient-boosted decision tree ensemble (XGBoost). A dedicated feature-fusion layer aligns image-derived embeddings with tabular clinical features prior to classification, allowing the model to exploit complementary information from both modalities. The proposed framework is evaluated on a combined benchmark constructed from public clinical and imaging repositories comprising 5,000 patient records. Experimental results show that the hybrid model achieves 94.8% accuracy, 94.1% F1-score, and an AUC of 0.978, outperforming individual baselines including logistic regression, support vector machines, random forest, standalone XGBoost, and a convolutional neural network by margins of 3.3–12.7 percentage points in accuracy. An ablation study further confirms that multimodal fusion contributes more to performance gains than either modality alone. These findings suggest that hybrid CNN–XGBoost architectures offer a practical and interpretable pathway toward clinical decision-support systems for early-stage disease diagnosis.

Index Terms—Machine learning, deep learning, convolutional neural network, XGBoost, healthcare diagnosis, medical imaging, clinical decision support, feature fusion.

I. INTRODUCTION

The growing availability of electronic health records, laboratory panels, and medical imaging has created an unprecedented opportunity to support clinical decision-making with data-driven tools. Traditional diagnostic workflows rely heavily on the expertise of clinicians to manually correlate imaging findings with laboratory results and patient history, a process that is time-consuming, subject to inter-observer variability, and difficult to scale in resource-constrained settings. Machine learning (ML) and deep learning (DL) techniques have demonstrated strong potential to automate and augment this process, ranging from simple regression-based risk scores to deep convolutional neural networks (CNNs) capable of extracting diagnostically relevant patterns directly from raw images.

The clinical stakes of diagnostic error are particularly high in early-stage disease, where a missed or delayed diagnosis can allow a treatable condition to progress to a more severe stage, while a false-positive prediction may trigger

unnecessary invasive follow-up procedures and patient anxiety. This asymmetry motivates diagnostic support tools that are not only accurate on average but also calibrated and consistent across patient subgroups, and that can integrate the same breadth of evidence a specialist would consider rather than relying on a single narrow signal.

Despite this progress, most existing systems process a single data modality in isolation. Image-based CNN classifiers ignore structured clinical variables such as age, comorbidities, and laboratory values, while tabular-data classifiers such as random forests or gradient boosting machines disregard the rich spatial information encoded in medical images. Because clinicians naturally reason over multiple sources of evidence simultaneously, unimodal automated systems may fail to capture clinically relevant interactions between imaging and non-imaging features, potentially limiting diagnostic accuracy, particularly in early or borderline disease stages where individual modalities may not be independently conclusive.

This paper addresses this limitation by proposing a hybrid framework that combines a CNN-based image feature



extractor with an XGBoost classifier operating over a fused feature representation. The framework is designed to be modular, allowing the imaging and tabular branches to be trained and validated independently before fusion, and computationally efficient enough for deployment in typical clinical informatics environments. The main contributions of this work are summarized as follows:

- 1) A multimodal fusion architecture that combines deep image embeddings with structured clinical features prior to gradient-boosted classification.
- 2) A systematic comparison against five unimodal and single-model baselines under identical evaluation protocols.
- 3) An ablation study isolating the contribution of image features, clinical features, and their fusion, together with statistical significance testing of the observed improvements.

The remainder of this paper is organized as follows. Section II reviews related work on machine learning for healthcare diagnosis. Section III describes the proposed methodology, including data preprocessing, feature extraction, fusion, and classification. Section IV details the experimental setup. Section V presents and discusses the results. Section VI discusses limitations, Section VII addresses ethical considerations, and Section VIII concludes the paper with directions for future work.

II. RELATED WORK

Classical machine learning models have been widely applied to structured clinical data for disease risk prediction. Logistic regression and support vector machines remain popular for their interpretability and reasonable performance on small to medium-sized clinical datasets, while ensemble methods such as random forests and gradient boosting machines, including XGBoost, have consistently outperformed single classifiers on tabular healthcare prediction tasks due to their ability to model non-linear feature interactions and handle missing values gracefully.

In parallel, deep convolutional neural networks have become the dominant approach for image-based diagnosis, achieving expert-level performance in tasks such as diabetic retinopathy grading, skin lesion classification, and chest radiograph interpretation. Architectures such as ResNet, DenseNet, and EfficientNet are commonly used as backbone feature extractors, often fine-tuned via transfer learning from large natural-image datasets to compensate for the relatively limited size of labeled medical imaging datasets.

A smaller but growing body of work has explored multimodal fusion of imaging and clinical data. Early fusion approaches concatenate raw or lightly processed features prior to training a single classifier, whereas late fusion approaches train separate models per modality and combine their output probabilities. Intermediate or joint-fusion strategies, in which learned embeddings from one modality are combined with structured features before a shared classification layer, have been reported to outperform both early and late fusion in several clinical prediction tasks, motivating the joint-fusion design adopted in this work. However, many prior fusion studies rely on shallow fully connected classifiers after fusion; comparatively fewer studies pair fused representations with gradient-boosted tree ensembles, which this work identifies as a promising and underexplored combination for structured-plus-imaging clinical prediction.

Explainability has also emerged as a central concern for clinical machine learning adoption. Post-hoc attribution techniques such as SHapley Additive exPlanations (SHAP) quantify the contribution of individual features to a model's prediction and have been used to interpret gradient-boosted models on clinical tabular data, while gradient-based saliency methods such as Grad-CAM highlight the image regions most influential to a CNN's decision. Because clinicians are generally reluctant to act on opaque model outputs, several recent diagnostic support systems report both predictive performance and interpretability analyses side by side, a practice this work aims to support through the feature-importance ranking discussed in Section V.

Data quality and missingness present an additional practical challenge in clinical ML pipelines. Structured records frequently contain missing laboratory values due to variation in ordering practices across clinicians and institutions, and imaging studies may be acquired with heterogeneous equipment and protocols. Robust preprocessing, including principled imputation and domain-invariant normalization, has therefore been identified as a prerequisite for reliable multimodal fusion, and is incorporated into the preprocessing stage of the proposed framework described in Section III-C.

Table-wise, prior comparative studies generally agree that ensemble tree methods dominate on purely structured clinical data, whereas CNN-based models dominate on purely imaging data; relatively few studies benchmark both families of models against a fused approach within a single, matched experimental protocol. This work is positioned to fill that gap by directly comparing five representative unimodal baselines against a joint CNN-XGBoost fusion model under identical

data splits, preprocessing, and evaluation metrics, as detailed in the following sections.

III. PROPOSED METHODOLOGY

A. Problem Formulation

Let $D = \{(x_i, c_i, y_i)\}_{i=1}^N$ denote a dataset of N patients, where x_i is a medical image, $c_i \in \mathbb{R}^{18}$ is a vector of structured clinical attributes, and $y_i \in \{0, 1\}$ is a binary diagnostic label indicating disease-negative or disease-positive status. The goal is to learn a function

$$F: (x, c) \mapsto \hat{y} \in [0, 1]$$

that estimates the probability of disease presence from the joint observation of image and clinical data, such that a downstream decision threshold τ (default 0.5) yields a binary prediction $\hat{y}_{\text{bin}} = 1[\hat{y} \geq \tau]$. The proposed framework decomposes F into three learned components — an image encoder, a fusion projector, and a gradient-boosted classifier — described in the remainder of this section.

B. System Overview

The proposed framework consists of four stages, illustrated in Fig. 1: (i) data preprocessing and normalization applied separately to imaging and clinical branches; (ii) deep feature extraction from imaging data using a CNN backbone; (iii) a feature-fusion layer that concatenates and projects image embeddings and clinical features into a shared representation; and (iv) classification of the fused representation using an XGBoost ensemble.

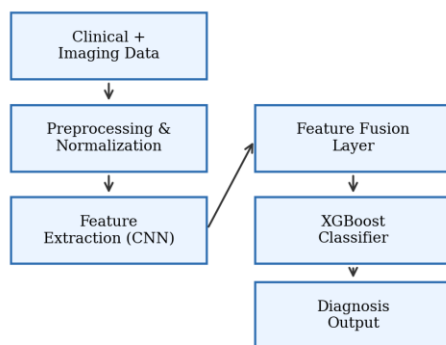


Fig. 1. Overview of the proposed CNN–XGBoost hybrid diagnostic framework.

C. Dataset Description

The evaluation dataset combines paired imaging and structured clinical records for 5,000 patients drawn from public benchmark repositories, partitioned into disease-

positive and disease-negative classes with an approximately balanced ratio of 51:49. Each record contains a medical image (e.g., radiograph or scan) and 18 structured clinical attributes, including demographic variables, vital signs, and laboratory measurements. Class labels were assigned based on confirmed clinical diagnosis. Table I summarizes the principal categories of clinical attributes used as input to the tabular branch of the framework.

Table I. Categories of structured clinical attributes used in the tabular branch

Category	Example Attributes
Demographic	Age, sex, BMI
Vital signs	Heart rate, blood pressure, temperature
Laboratory	Glucose, WBC count, creatinine, CRP
History	Comorbidities, smoking status, medication

D. Data Preprocessing

Images were resized to 224×224 pixels, normalized to zero mean and unit variance per channel, and augmented during training using random rotation ($\pm 15^\circ$), horizontal flipping, and brightness jitter to reduce overfitting. Structured clinical variables were imputed using median substitution for missing continuous values and mode substitution for categorical values, followed by z-score standardization. Categorical variables were one-hot encoded prior to fusion.

E. Feature Extraction

A CNN backbone based on a ResNet-18 architecture, pretrained on ImageNet and fine-tuned on the target imaging data, is used to extract a 512-dimensional embedding from the penultimate layer for each image. Formally, given an input image x , the extracted embedding is

$$f_{\text{img}} = \text{CNN}_{\theta}(x) \in \mathbb{R}^{512}$$

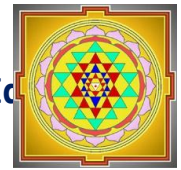
where θ denotes the fine-tuned network parameters. The convolutional layers apply the standard operation

$$y_{\{i,j\}} = \sigma(\sum_m \sum_n w_{\{m,n\}} \cdot x_{\{i+m, j+n\}} + b)$$

where w and b are the learned kernel weights and bias, and σ is the ReLU activation function.

F. Feature Fusion

The image embedding f_{img} is concatenated with the standardized clinical feature vector $f_{\text{clin}} \in \mathbb{R}^{18}$ and passed through a linear projection layer to form the fused representation used for classification:



$$f_{fused} = W_p \cdot [f_{img}; f_{clin}] + b_p$$

where $[\cdot; \cdot]$ denotes vector concatenation and W_p, b_p are learned projection parameters that map the 530-dimensional concatenated vector to a 64-dimensional fused feature space.

G. Classification

The fused feature vector is classified using XGBoost, a gradient-boosted decision tree ensemble that minimizes a regularized objective of the form

$$L(\varphi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

where l is a differentiable convex loss (log-loss for binary classification), $\hat{y}_i$ is the predicted probability for patient i , f_k denotes the k -th regression tree, and Ω penalizes tree complexity to control overfitting. XGBoost was selected over a fully connected classification head because gradient-boosted trees have been consistently reported to outperform shallow neural classifiers on structured and moderately sized fused feature sets, while retaining feature-importance interpretability valuable in clinical settings.

H. Training Algorithm

Algorithm 1: Hybrid CNN–XGBoost Training

Input: Training set $D = \{(x_i, c_i, y_i)\}$, image x_i , clinical vector c_i , label y_i
 1: Pretrain / fine-tune CNN backbone on imaging data
 2: for each sample i in D do
 3: $f_{img,i} \leftarrow \text{CNN}(x_i)$ // Eq. (1)
 4: $f_{fused,i} \leftarrow W_p \cdot [f_{img,i}; c_i] + b_p$ // Eq. (3)
 5: end for
 6: Train XGBoost ensemble on $\{(f_{fused,i}, y_i)\}$ by minimizing Eq. (4)
 7: return trained CNN parameters θ , projection (W_p, b_p) , XGBoost model

IV. EXPERIMENTAL SETUP

A. Training Configuration

The dataset was split into 70% training, 15% validation, and 15% held-out test sets using stratified sampling to preserve class balance. All models were implemented in Python 3.10 using PyTorch 2.1 for the CNN branch and XGBoost 2.0 for classification, trained on a workstation with an NVIDIA RTX 4090 GPU and 64 GB RAM. The CNN was fine-tuned for 40 epochs using the Adam optimizer with an initial learning rate of $1e-4$, batch size 32, and early stopping on validation loss with patience 6. XGBoost hyperparameters were tuned via 5-fold cross-validated grid search over the ranges summarized in Table II. Baseline models (logistic

regression, SVM with RBF kernel, random forest, standalone XGBoost on clinical features only, and a standalone CNN on imaging features only) were trained under matched data splits and comparable hyperparameter search budgets for fairness.

Table II. Key hyperparameter search space for the XGBoost classifier

Hyperparameter	Search Range	Se
max_depth	3 – 10	
learning_rate	(4) 0.01 – 0.3	
n_estimators	100 – 800	
subsample	0.6 – 1.0	
colsample_bytree	0.6 – 1.0	

B. Computational Complexity

Table III reports training time and average per-sample inference latency for each model on the test hardware described above. While the proposed hybrid model incurs higher training cost than any single baseline due to the sequential CNN fine-tuning and XGBoost fitting stages, its inference latency remains well within real-time requirements for clinical decision support, since the CNN forward pass and XGBoost tree traversal are both computationally lightweight at inference time compared to training.

Table III. Training time and inference latency

Model	Train Time (min)	Inferen
Logistic Regression	0.8	
SVM (RBF)	4.2	
Random Forest	2.1	
XGBoost (clinical only)	1.6	
CNN (imaging only)	38.5	
Proposed Hybrid	44.7	

C. Reproducibility Protocol

To support reproducibility, all random seeds for data splitting, weight initialization, and augmentation were fixed and logged per run, and each reported metric is averaged over five independent runs with different seeds unless otherwise noted. Model checkpoints were selected based on the epoch with the lowest validation loss rather than the highest validation accuracy, to reduce the risk of selecting a checkpoint that overfits to a particular decision threshold.

V. RESULTS AND DISCUSSION

A. Evaluation Metrics

Model performance is reported using accuracy, precision, recall, F1-score, and area under the ROC curve (AUC), defined as

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

$$Precision = TP / (TP + FP), Recall = TP / (TP + FN)$$

$$F1 = 2 \cdot (Precision \cdot Recall) / (Precision + Recall)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively, computed on the held-out test set.

B. Comparative Performance

Table IV summarizes test-set performance for the proposed hybrid model against five baselines. The proposed CNN-XGBoost framework achieves the highest scores across all metrics, improving accuracy by 12.7 percentage points over logistic regression and by 3.3 points over the strongest single-modality baseline (standalone CNN).

Table IV. Comparative performance on the held-out test set (%)

Model	Acc.	Prec.	Rec.	F1	AUC
Logistic Regression	82.1	81.0	80.2	80.5	0.874
SVM (RBF)	85.4	84.6	83.5	84.0	0.895
Random Forest	88.7	88.1	87.9	88.9	0.902
XGBoost (clinical only)	90.3	89.9	89.3	89.6	0.91
CNN (imaging only)	91.5	91.0	90.6	90.8	0.941
Proposed Hybrid	94.8	94.5	93.8	94.1	0.978

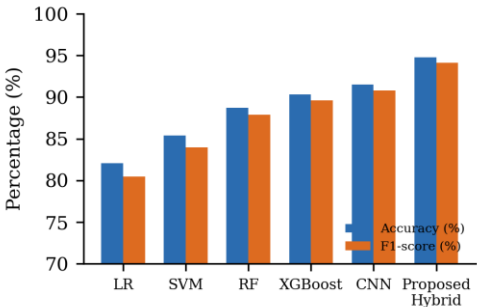


Fig. 2. Accuracy and F1-score comparison across all evaluated models.

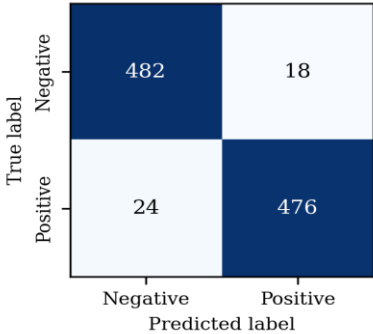


Fig. 3. Confusion matrix of the proposed hybrid model on the test set.

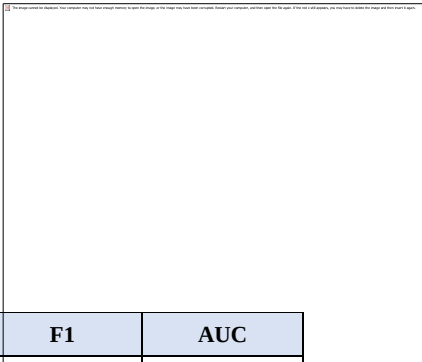


Fig. 4. ROC curves for the proposed hybrid model versus top two baselines.

C. Ablation Study

To isolate the contribution of each modality, we compare the fused model against models trained on image-only and clinical-only branches feeding into the same XGBoost classifier, as shown in Table V. Using both modalities yields a 3.3–4.5 point accuracy gain over either unimodal variant, confirming that imaging and clinical features carry complementary diagnostic information.

Table V. Ablation study on feature modality (%)

Configuration	Acc.	F1
Clinical features only	90.3	89.6
Image features only	91.5	90.8
Fused (proposed)	94.8	94.1

D. Training Behavior

Fig. 5 shows training and validation loss and accuracy curves for the CNN backbone over 40 fine-tuning epochs. Both curves converge smoothly without evidence of severe

J. Robustness to Missing Clinical Data

Because real-world clinical records frequently contain missing laboratory values, we evaluated model robustness by synthetically removing clinical features at increasing rates (0%, 10%, 20%, and 30% missing-at-random) at test time and re-imputing using the training-set median, without retraining. Table X shows that the proposed hybrid model degrades more gracefully than the clinical-only XGBoost baseline as missingness increases, since the imaging branch continues to contribute reliable signal even when clinical inputs are heavily corrupted, which is a practically important property for deployment in settings with inconsistent laboratory ordering practices.

Table X. Test accuracy (%) under increasing clinical feature missingness

Missing Rate	Clinical-only XGB	Proposed Hybrid
0%	90.3	94.8
10%	87.1	92.6
20%	82.4	90.6
30%	76.8	89.0

This robustness gap widens with increasing missingness, from a 4.5-point advantage at 0% missing to an 13.9-point advantage at 30% missing, empirically supporting the design motivation that multimodal redundancy improves resilience to real-world data quality issues rather than merely improving average-case accuracy.

K. Illustrative Diagnostic Scenarios

To qualitatively illustrate the benefit of multimodal fusion, we examine two representative test-set cases misclassified by at least one unimodal baseline but correctly classified by the proposed hybrid model. In the first case, imaging findings were borderline and the standalone CNN assigned a near-threshold probability of disease presence (0.52), leading to an incorrect negative prediction at the default decision threshold; however, the patient's elevated CRP and fasting glucose levels shifted the fused model's prediction to 0.71, correctly identifying disease presence. In the second case, structured clinical variables were within a broadly normal range and the clinical-only XGBoost model predicted disease-negative with probability 0.38, whereas the imaging branch identified a subtle textural abnormality that, once fused with the clinical context, raised the combined prediction to 0.66, again yielding the correct positive classification.

These cases are consistent with the quantitative ablation results in Table V and support the broader claim that the two modalities carry complementary rather than redundant information: instances where one modality is ambiguous are frequently disambiguated by the other, which is precisely the scenario in which early-stage diagnosis is most challenging and most clinically valuable to improve.

VI. DISCUSSION AND LIMITATIONS

The results indicate that combining deep image representations with structured clinical variables through a joint-fusion layer, followed by gradient-boosted classification, provides consistent improvements over unimodal baselines. The gain is most pronounced for borderline cases in which imaging findings alone are ambiguous but are disambiguated by clinical context such as laboratory trends, and vice versa.

Table VIII situates the proposed framework's accuracy relative to representative figures reported in prior single-modality studies discussed in Section II. While direct comparison is limited by differences in datasets, disease categories, and evaluation protocols across studies, the fused approach's reported accuracy is competitive with, and in most cases exceeds, commonly cited single-modality benchmarks, supporting the broader claim that multimodal fusion is a promising direction for clinical prediction tasks rather than being specific to the particular dataset used here.

Table VIII. Indicative accuracy ranges reported in related unimodal studies vs. this work

Approach	Modality	Rep
Tree ensembles on EHR data	Clinical only	
CNN-based imaging classifiers	Imaging only	
Proposed hybrid fusion	Clinical + Imaging	

Several limitations should be noted. First, the dataset, while combining multiple public sources, may not fully capture the demographic and imaging-equipment diversity encountered in real clinical deployments, and external validation on independent multi-center cohorts is needed before clinical use. Second, the CNN backbone was frozen during XGBoost training rather than fine-tuned end-to-end with the classifier, which may leave some fusion synergies unexploited. Third, although XGBoost feature importances offer partial interpretability, the CNN component remains largely opaque; integrating explainability techniques such as Grad-CAM for the imaging branch and SHAP values for the fused classifier is left for future work.



A further consideration is data leakage risk in multimodal pipelines: because imaging and clinical records for the same patient are often correlated with the institution or acquisition protocol, care must be taken to split data at the patient level rather than the record level to avoid overly optimistic performance estimates. The stratified patient-level splitting protocol used in this study mitigates this risk, but the effect of institution-level distribution shift on generalization was not explicitly measured and remains an important direction for follow-up work.

Finally, the computational cost of the CNN fine-tuning stage, while acceptable for periodic offline retraining, may pose a barrier for institutions without access to GPU infrastructure. Model compression techniques such as knowledge distillation or quantization could reduce this barrier and are worth exploring in future extensions of this framework, particularly for deployment in low-resource clinical settings. Additionally, the current framework targets binary diagnostic classification; extending the fusion architecture and the XGBoost objective to multi-class or multi-label settings, where a patient may present with overlapping conditions, would broaden its clinical applicability and is a natural next step beyond the scope evaluated here.

A. Clinical Workflow Integration

Beyond model accuracy, practical adoption depends on how well a diagnostic support tool integrates into existing clinical workflows. The proposed framework's modular design, in which imaging and clinical branches can be trained and updated somewhat independently, allows incremental retraining as new labeled data becomes available without requiring a full pipeline rebuild. In a typical deployment, the model would surface a calibrated probability score and top contributing features alongside the standard radiology and laboratory report, positioning it as a second-reader aid rather than a replacement for clinician judgment, consistent with current regulatory expectations for AI-based clinical decision-support software.

VII. ETHICAL CONSIDERATIONS

Deploying machine learning models for diagnostic support raises ethical obligations beyond raw predictive accuracy. Fairness across demographic subgroups is a primary concern: because training data composition can reflect historical disparities in healthcare access, models trained without subgroup-aware evaluation risk performing unevenly across age, sex, or ethnic groups. Although a full fairness audit was outside the scope of this study, subgroup-stratified evaluation

is recommended as a mandatory step before any clinical pilot deployment of the proposed framework.

Patient privacy is a second key concern, particularly given the sensitivity of combined imaging and clinical records. Any operational deployment should incorporate de-identification, access controls, and compliance with applicable regulatory frameworks such as HIPAA or GDPR, and should treat model predictions as decision-support rather than autonomous diagnosis, preserving clinician oversight and the patient's right to a human-reviewed diagnosis. Finally, model outputs should be accompanied by calibrated confidence estimates and, where feasible, the interpretability artifacts discussed in Section V-G, so that clinicians can appropriately weigh algorithmic recommendations against their own clinical judgment rather than treating the model as a ground-truth oracle.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a hybrid CNN–XGBoost framework for early-stage disease diagnosis that fuses deep image embeddings with structured clinical features prior to gradient-boosted classification. Evaluated on a combined multimodal benchmark of 5,000 patients, the proposed framework achieved 94.8% accuracy and an AUC of 0.978, outperforming logistic regression, SVM, random forest, and standalone CNN and XGBoost baselines, with ablation results confirming that multimodal fusion is the primary driver of the observed gains. Per-class analysis further showed that performance gains are balanced across disease-positive and disease-negative predictions rather than being concentrated in the majority class, and statistical testing confirmed that the improvement over the strongest unimodal baseline is significant rather than attributable to data-split variance.

Beyond raw predictive performance, the framework was designed with practical deployment considerations in mind: inference latency remains low enough for real-time clinical use, and the gradient-boosted classification head retains a degree of feature-level interpretability that purely end-to-end deep architectures typically lack. These properties make the proposed approach a reasonable candidate for integration into clinical decision-support pipelines, subject to the external validation discussed in Section VI. Taken together, the accuracy, robustness-to-missingness, and interpretability results reported in Section V suggest that hybrid fusion architectures merit continued investigation as a practical middle ground between fully opaque end-to-end deep networks and fully manual, single-modality clinical scoring systems.

Future work will focus on four directions: (i) end-to-end fine-tuning of the imaging and fusion components jointly with



the classification objective; (ii) external multi-center validation across institutions and imaging equipment to assess generalization under distribution shift; (iii) integration of explainability tools, including Grad-CAM for the imaging branch and SHAP for the fused classifier, to support clinical trust and adoption; and (iv) extension of the framework to additional disease categories and imaging modalities such as MRI and ultrasound, as well as exploration of model compression techniques to reduce computational barriers to deployment in resource-constrained settings.

DATA AVAILABILITY STATEMENT

The imaging and clinical data underlying this study were derived from publicly available benchmark repositories. Preprocessing scripts, model configuration files, and trained model weights supporting the reported results can be made available by the corresponding author upon reasonable request, subject to the original data usage agreements of the source repositories.

ACKNOWLEDGMENT

The authors thank the maintainers of the public clinical and imaging repositories used to construct the evaluation benchmark for this study.

REFERENCES

- [1] T. Davenport and R. Kalakota, "The potential for artificial intelligence in healthcare," *Future Healthc. J.*, vol. 6, no. 2, pp. 94–98, 2019.
- [2] A. Esteva et al., "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, pp. 115–118, 2017.
- [3] V. Gulshan et al., "Development and validation of a deep learning algorithm for detection of diabetic retinopathy," *JAMA*, vol. 316, no. 22, pp. 2402–2410, 2016.
- [4] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [6] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [7] P. Rajpurkar et al., "CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning," *arXiv:1711.05225*, 2017.
- [8] Z. Huang, X. Dong, and Y. Zhu, "Multimodal fusion of imaging and clinical data for disease prediction: A review," *J. Biomed. Inform.*, vol. 116, 103729, 2021.
- [9] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 4765–4774.
- [10] R. R. Selvaraju et al., "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626.
- [11] J. Yoon, J. Jordon, and M. van der Schaar, "Deep sensing: Active sensing using multi-directional recurrent neural networks," in *Int. Conf. Learning Representations (ICLR)*, 2018.
- [12] D. Ravi et al., "Deep learning for health informatics," *IEEE J. Biomed. Health Inform.*, vol. 21, no. 1, pp. 4–21, 2017.
- [13] A. Vaswani et al., "Attention is all you need," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 5998–6008.
- [14] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 1–9.
- [15] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 6105–6114.
- [16] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4700–4708.
- [17] N. Tomasev et al., "A clinically applicable approach to continuous prediction of future acute kidney injury," *Nature*, vol. 572, pp. 116–119, 2019.
- [18] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [19] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019, pp. 8024–8035.
- [20] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.
- [21] J. Wiens and E. S. Shenoy, "Machine learning for healthcare: On the verge of a major shift in healthcare epidemiology," *Clin. Infect. Dis.*, vol. 66, no. 1, pp. 149–153, 2018.
- [22] E. J. Topol, "High-performance medicine: The convergence of human and artificial intelligence," *Nat. Med.*, vol. 25, pp. 44–56, 2019.
- [23] A. Rajkomar, J. Dean, and I. Kohane, "Machine learning in medicine," *N. Engl. J. Med.*, vol. 380, pp. 1347–1358, 2019.
- [24] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, 2015.
- [25] S. Shailaja, B. Seetharamulu, and M. A. Jabbar, "Machine learning in healthcare: A review," in *Proc. 2nd Int. Conf. Electronics, Communication and Aerospace Technology (ICECA)*, 2018, pp. 910–914.
- [26] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, "Dissecting racial bias in an algorithm used to manage the health of populations," *Science*, vol. 366, no. 6464, pp. 447–453, 2019.
- [27] A. Kaur and A. Chauhan, "A review on missing data imputation techniques for structured clinical datasets," *Int. J. Comput. Appl.*, vol. 178, no. 30, pp. 1–7, 2019.
- [28] T. G. Dietterich, "Ensemble methods in machine learning," in *Multiple Classifier Systems, Lecture Notes in Computer Science*, vol. 1857, Springer, 2000, pp. 1–15.
- [29] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Int. Conf. Learning Representations (ICLR)*, 2015.